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Roll No

EE-5001-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Utilisation of Electrical Energy

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions,
ii) All questions carry equal marks.

1. a) Explain the laws of Illumination. 7
b) A lamp of 500 watts having mscp of 1000 is suspended 2.7 meters above the working plane. Calculate: 7
 - i) Illumination directly below the lamp at the working plane
 - ii) Lamp efficiency
 - iii) Illumination at a point 2.6 meters away on the horizontal plane from vertically below the lamp.
2. a) List the advantages of electric heating. 7
b) Explain the induction heating operating principle and the application and limitation and above. 7
3. a) Discuss the principle of arc welding and differentiate carbon and metallic arc welding. 7
b) What is electroplating and what for is it done? Describe the various operations involved in electroplating. 7

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4. a) Give different system of electric traction and their advantages and disadvantages. 7
b) A train runs with average speed of 60 kmph. Distance between station is 3 kms, values of acceleration and retardation are 1.5 kmphs and 3.5 kmphs respectively. Find the maximum speed of train assuming trapezoidal speed time curve. 7
5. a) Explain different types of braking. 7
b) Draw and explain 25kV ac single phase traction systems. 7
6. a) Explain in brief the selection of motor drive for various industrial applications. 7
b) What are the advantages of electric drive over all other drives? 7
7. Explain the method of calculation of electrical loads for: 14
i) Refrigeration
ii) Air conditioning
8. Write short notes on (any two) 2×7=14
i) Load equalization
ii) Control of Traction motors
iii) Flood lightning
iv) Welding Transformers
